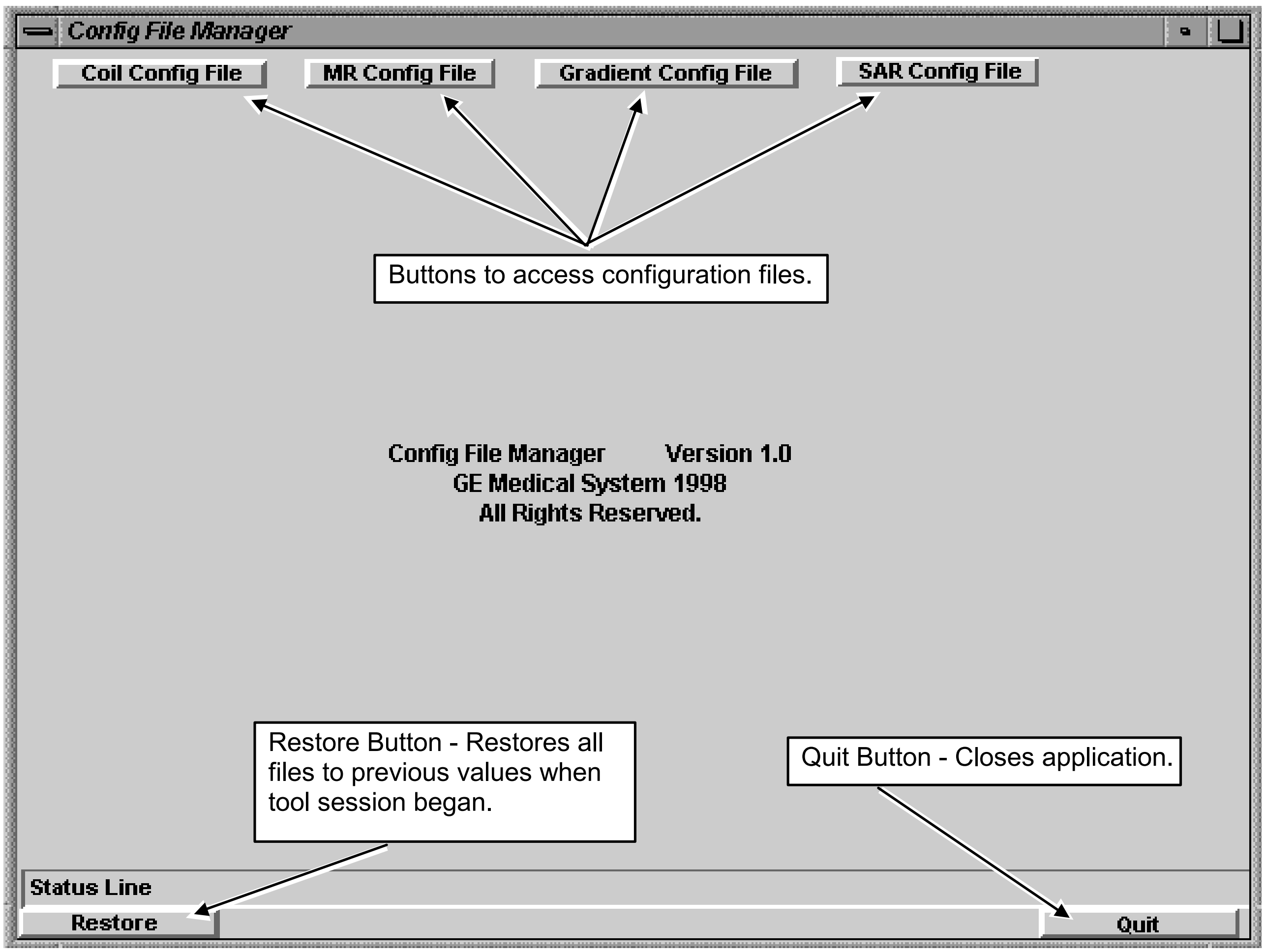
TABLE OF CONTENTS

1- OVERVIEW

The Configuration File Manager is a graphical interface used to change the contents of the Coil Configuration File, MR Configuration File, Gradient Configuration Files (there are two for ***TwinSpeed***), and the SAR Configuration File. The Configuration files customize the software to the present hardware configuration. (In this text, *configuration* appears as *config*.) It also contains pre-configured parameters for all GE coils, so adding a coil is as simple as "click to add"!

2- CONFIG FILE MANAGER MAIN SCREEN

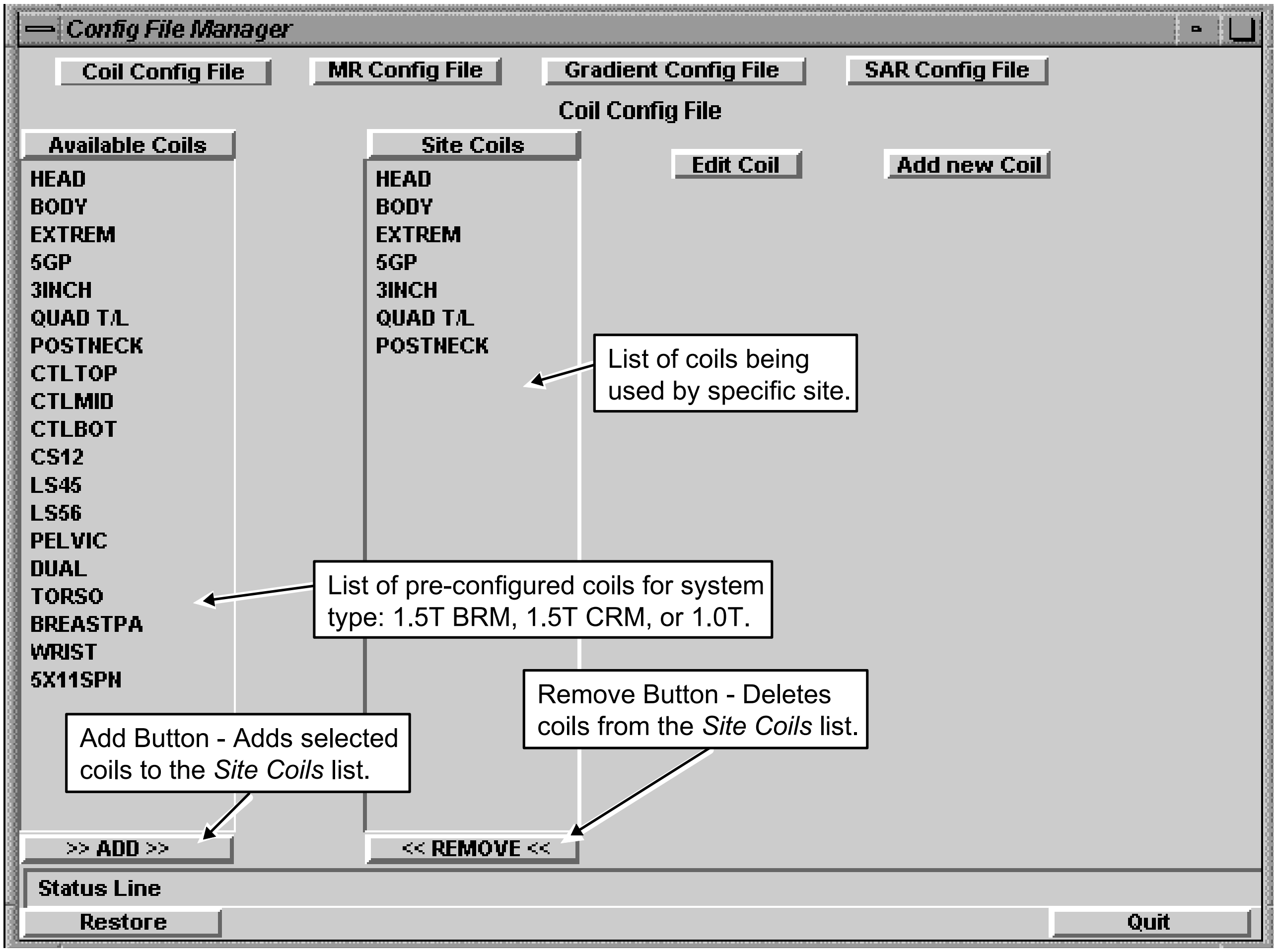
To start the Configuration File Manager, on the Service Desktop, select **[Utilities],** then **[Config File Manager]** and click **[Start].** The interface allows the user to make changes to the system configurations for the Coils, MR System, Gradients and SAR in one application. All of the default values are in these files. Options for editing and adding to the config files are available in each configuration category. See Illustration 2-1.



CONFIG FILE MANAGER MAIN SCREEN (TYPICAL)
ILLUSTRATION 2-1

3- COIL CONFIG FILE

The Coil Config File allows for the addition, removal and editing of coils being used by a specific site. See Illustration 3-1.



COIL CONFIG FILE SCREEN (TYPICAL)
ILLUSTRATION 3-1

The pre-configured coil lists for the Config File Manager tool are in Directory /w/config:

- CoilConfig.df.1.5T (1.5T with BRM Body Coil)
- CoilConfig.df.1.5T.crm (1.5T with CRM Body Coil)
- CoilConfig.df.1.0T (1.0T with BRM Body Coil)

Cable and coil loss values are different for each of these systems. During software install, the system copies the appropriate coil definition file above to the CoilConfig.df (in /w/config). The CoilConfig.df file is then used by the Config File Manager tool when adding an “Available Coil”.

Note

All coils available for Signa will be pre-configured as part of a software load. If no Restore Info is done to the system, It will be necessary for the installer to remove coils that are not being used by the site. Coils will only need to be added if the coils are not GE standard.

Note

For old revisions of software ASP, 8X, MEDRAD Transmit/Recieve Extremity Array Coil with configs LOCEXTPA, LGEXTPA, SMEXTPA will produce artifacts if autoshim is selected on anything but sagittal LOCEXTPA. A new CoilConfig.cfg value has been added to newer MR Software platforms for autoshimRcvr. This issue has been resolved with 9.0 and all SW revisions going forward.

IMPORTANT! **Do not multiply** a surface coil's Recon Scale Factor times the Head Coil Recon Scale Factor! Scan does this automatically. With the new Configuration File Manager, only the surface coil multiplier is pre-configured in the CoilConfig.cfg as the "Recon Scale Factor". When adding new coils, **only update the Recon Scale Factor for Head and Body coils**, depending on System Gain Calibration results.

3-1 Adding Coils

Note

All coils available for Signa will be pre-configured as part of a software load. If no Restore Info is done to the system, It will be necessary for the installer to remove coils that are not being used by the site. Coils will only need to be added if the coils are not GE standard.

To add coils to the Coil Config file, perform the following steps:

1. On the *Config File Manager* screen, select **[Coil Config File]**. See Illustration 2-1.
2. Select the coil(s) to add to the site's configuration by clicking on the desired coil name(s) in the *Available Coils* list. See Illustration 3-1.
3. Click the **[>>ADD>>]** button to add the selected coil(s) to the *Site Coils* list. You must reboot Signa for scan to recognize that a coil has been added to the list.

3-2 Deleting A Surface Coil

To delete coils from the Coil Config file, perform the following steps:

1. On the *Config File Manager* screen, select **[Coil Config File]**. See Illustration 2-1.
2. Select the coil(s) to delete in the *Site Coils* list by clicking on the coil name(s). See Illustration 3-1.
3. Click the **[<<Remove<<]** button to delete the selected coil(s) from the *Site Coils* list. You must reboot Signa for scan to recognize that a coil has been deleted from the list.

3-3 Editing Coil Values

To edit or view coil values perform the following steps:

1. On the *Config File Manager* screen, select **[Coil Config File]**. See Illustration 2-1.

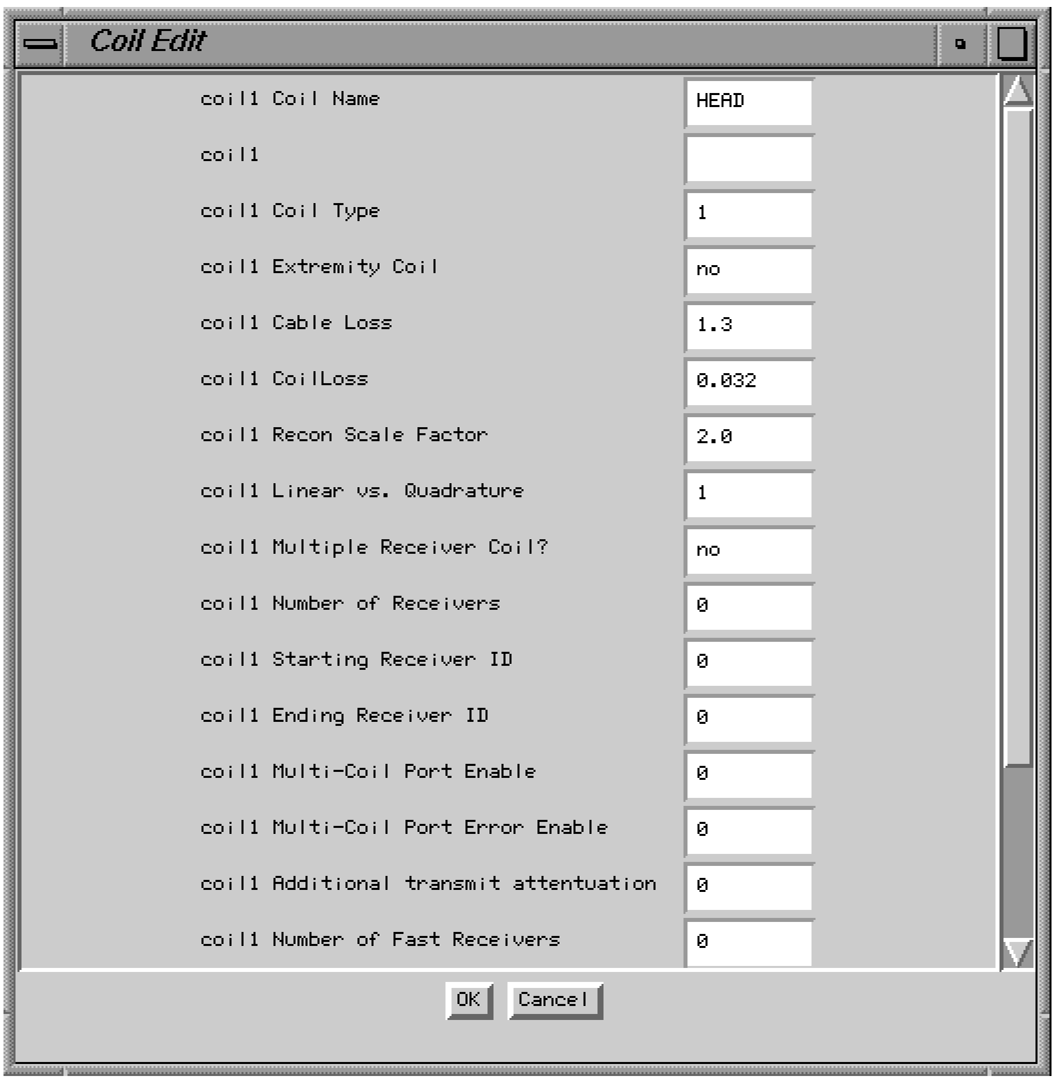
2. Select the appropriate coil(s) (you can select more than one coil) in the *Site Coils* list by clicking on the coil name(s) and then click the **[Edit Coil]** button. See Illustration 3-1.

Note:
Scan is sensitive to upper/lower case entries in the coil config file. Always use lower case **yes/no** entries for coil parameters (e.g. "Multiple Receiver coil?" and "Extremity Coil").

3. The *Coil Edit* screen displays the present values of the selected coils. See Illustration 3-2. To change a value, click in the field to enter text or tab through fields and make necessary changes. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located). Click **[OK]** to close the screen.

4. You must exit the tool and reboot Signa for scan to recognize coil configuration changes.

IMPORTANT! **Do not multiply** a surface coil's Recon Scale Factor times the Head Coil Recon Scale Factor! Scan does this automatically. Only the surface coil multiplier is pre-configured in the CoilConfig.cfg as the "Recon Scale Factor".



COIL EDIT SCREEN (TYPICAL)
ILLUSTRATION 3-2

3-4 Adding New Coils to Config File Manager

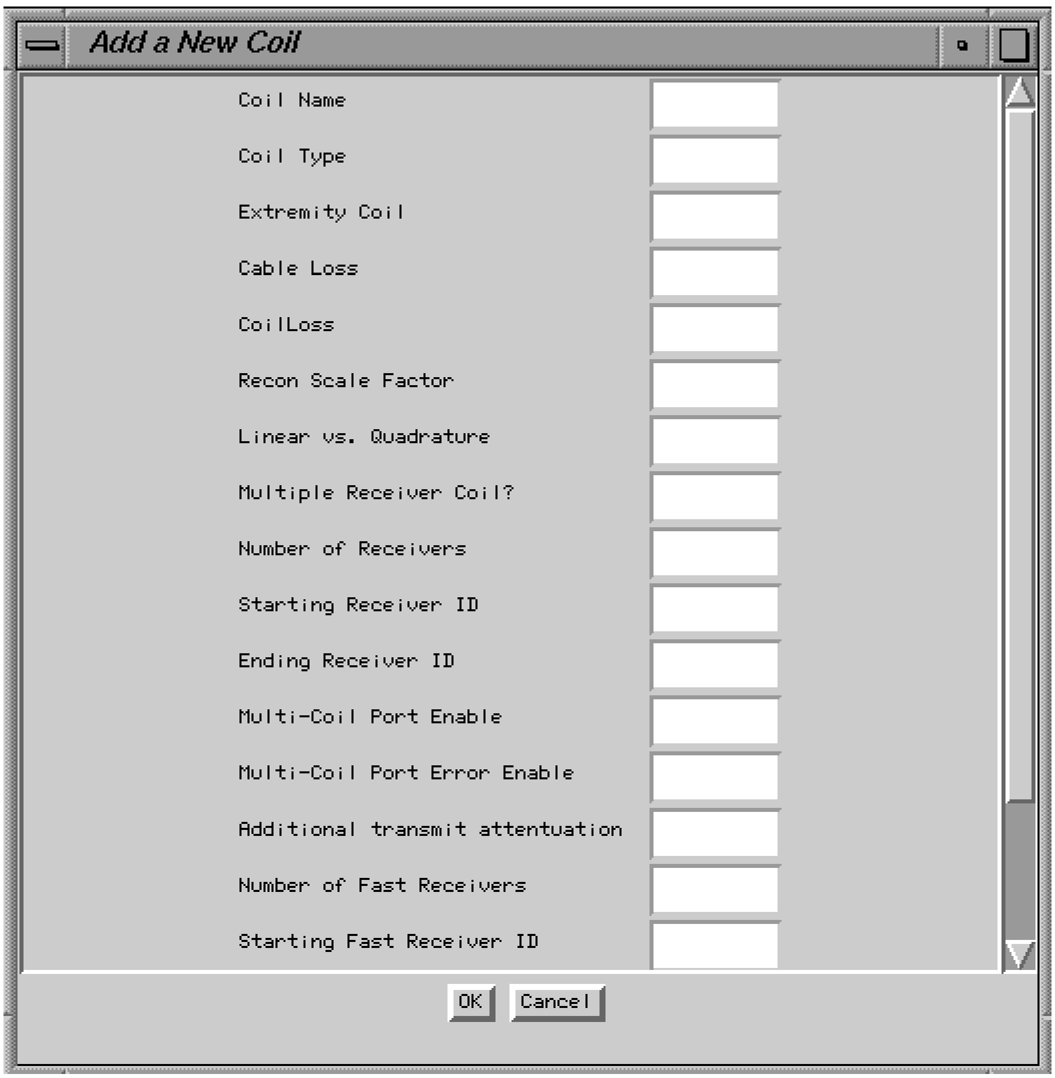
To add a coil that is not in the *Available Coils* list, perform the following steps:

1. On the *Config File Manager* screen, select **[Coil Config File]**. See Illustration 2-1.

2. Click the **[Add new Coil]** button. See Illustration 3-1.

3. The *Add a New Coil* screen displays coil value fields for a new coil. See Illustration 3-3. Values for new coils are in the service manual provided with the coil. For new "Multi Coil Recon Enable" field, enter **0** as the default unless instructed otherwise. Either click in the fields or tab through the fields to enter values. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located). Click **[OK]** to close the screen.

IMPORTANT! **Do not multiply** a surface coil's Recon Scale Factor times the Head Coil Recon Scale Factor! Scan does this automatically. With the new Configuration File Manager, the surface coil multiplier is pre-configured in the CoilConfig.cfg as the "Recon Scale Factor".



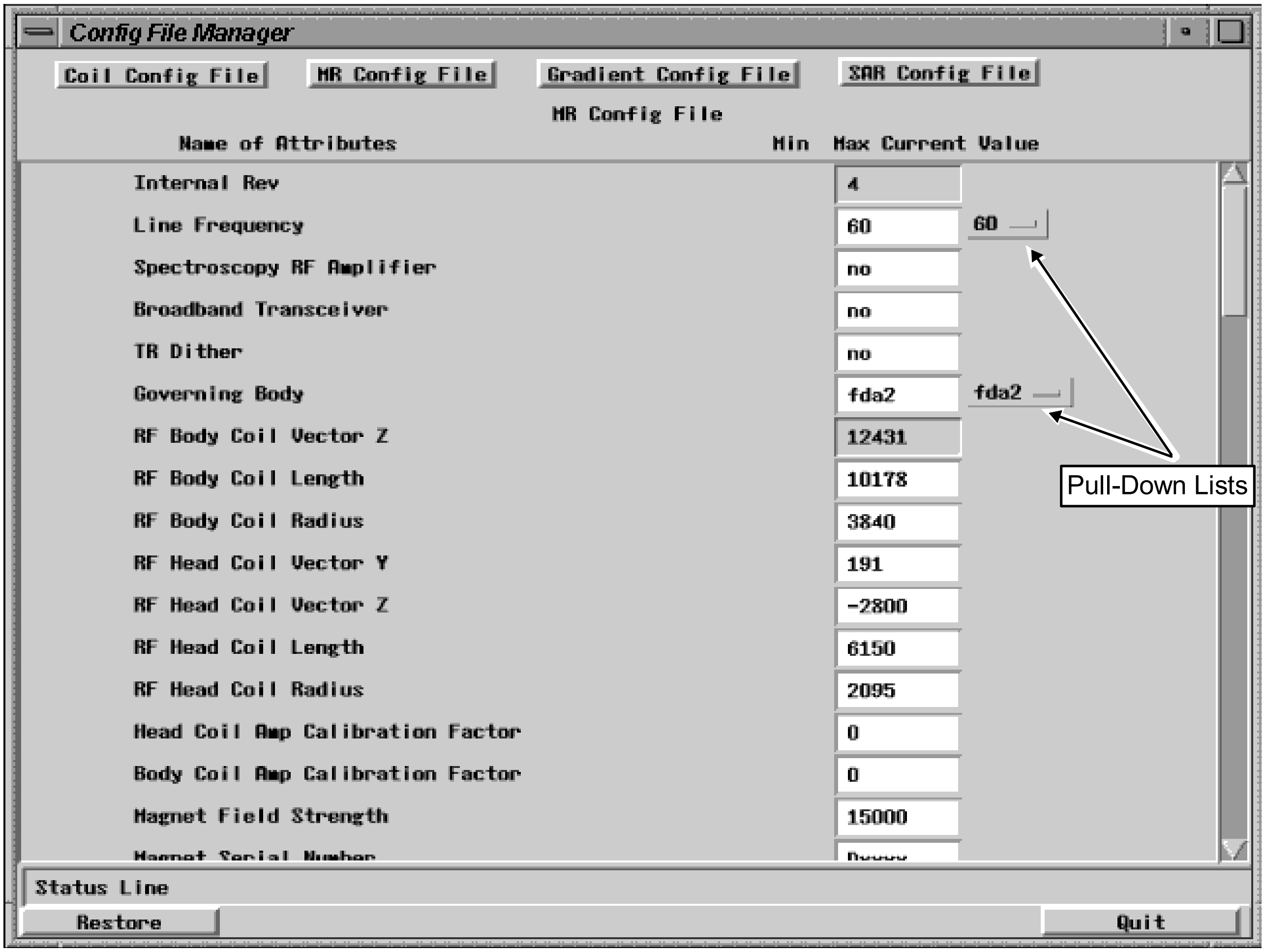
ADD NEW COIL SCREEN (TYPICAL)
ILLUSTRATION 3-3

4- MR CONFIG FILE

To edit and/or view the values of the MR Config File, perform the following steps:

1. On the *Config File Manager* screen, select **[MR Config File]**. See Illustration 2-1.

2. The *MR Config File* screen is displayed. See Illustration 4-1.



MR CONFIG FILE SCREEN (TYPICAL)
ILLUSTRATION 4-1

3. Either click in the fields or tab through the fields to enter values. For “Attributes” with defined values, click the pull-down list next to the current value. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located).

Note

Clicking in a field displays information about the selected field on the “Status Line”.

4. To apply changes made to the MR Config File, exit the config file manager. Make sure to click the **[Save]** button when the save dialog box appears. Then, shut down Signa and reboot the computer. (Refer to Section 7, *Exiting the Config File Manager*.)

5. See Appendix A for a listing of all the parameters in the MRconfig.cfg file.

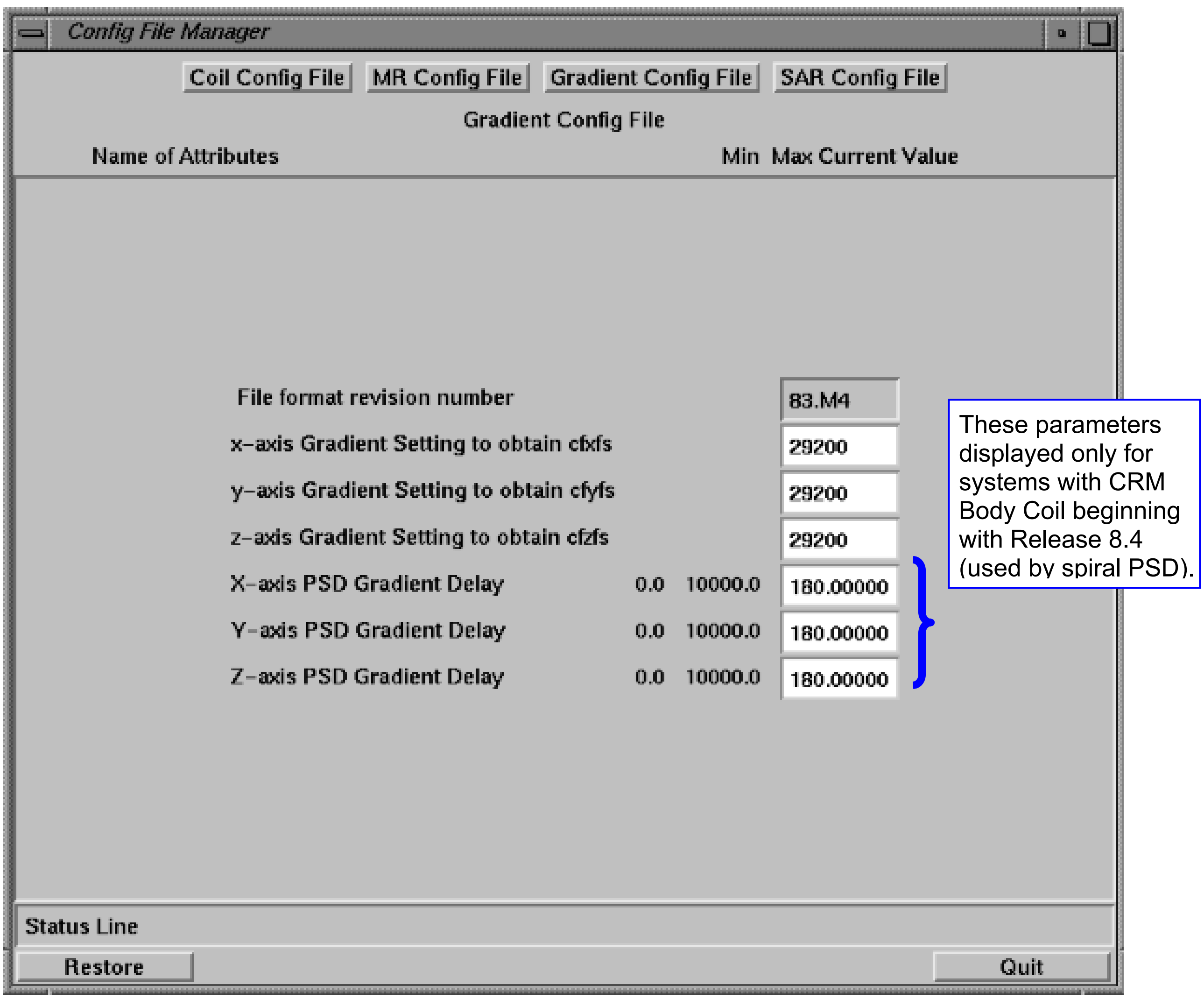
5- GRADIENT CONFIG FILE

To edit and/or view the values of the Gradient Config File, perform the following steps:

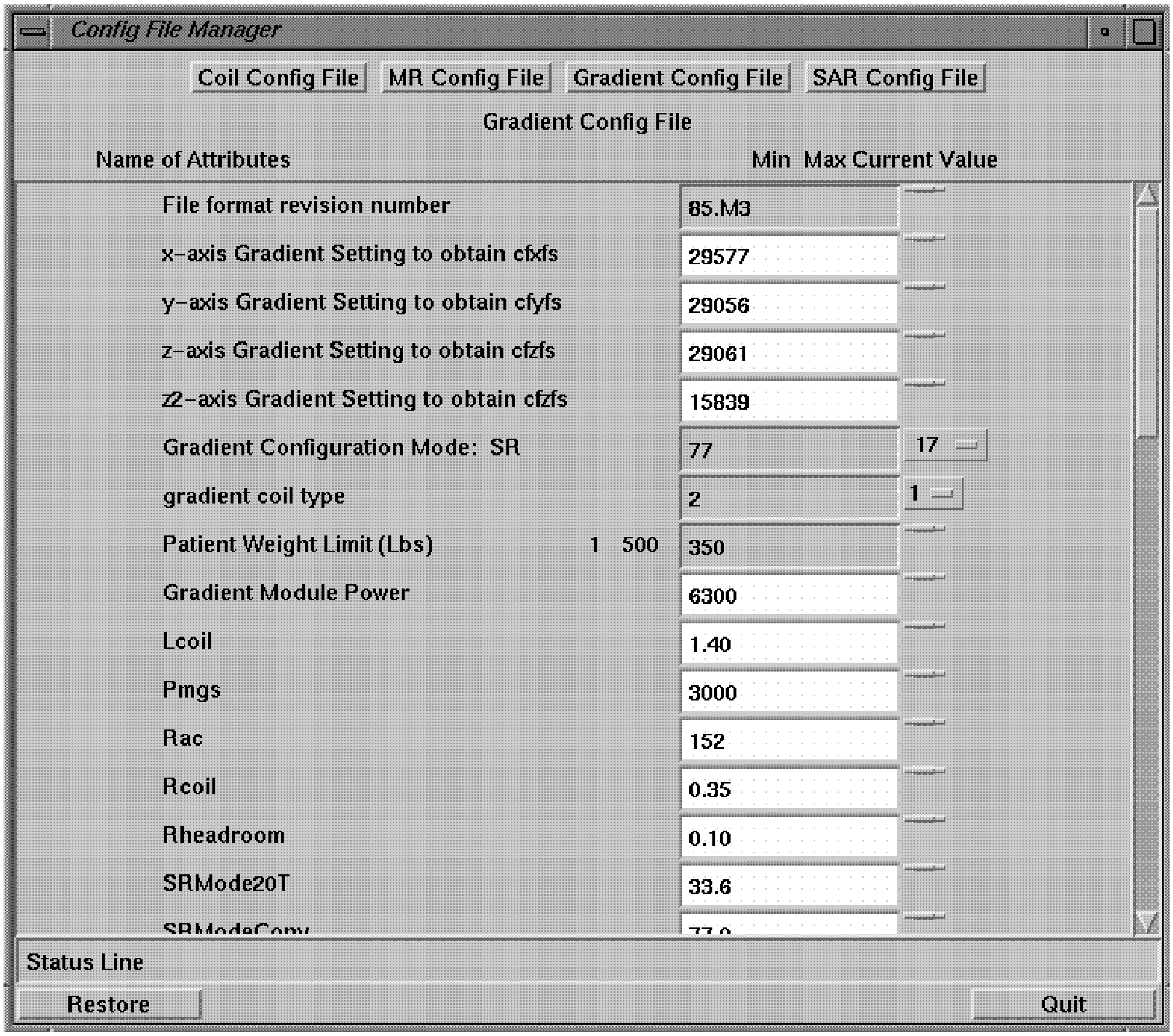
1. On the *Config File Manager* screen, select **[Gradient Config File]**. See Illustration 2-1.

2. The *Gradient Config File* screen is displayed. See Illustration 5-1 for Release 8.3/8.4 and Illustration 5-2 for Release ASP2.

For the ***TwinSpeed***, each **Gradient Mode**, Whole-Body (**WHOLE**) and Zoom (**ZOOM**), has a separate button.



GRADIENT CONFIG FILE SCREEN (RELEASE 8.3/8.4)
ILLUSTRATION 5-1



GRADIENT CONFIG FILE SCREEN (RELEASE 9.0 AND ASP2)
ILLUSTRATION 5-2

3. Either click in the fields or tab through the fields to enter. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located).

4. To apply changes made to the Gradient Config File, exit the config file manager. Make sure to click the **[Save]** button when the save dialog box appears. Then, shut down Signa and reboot the computer. (Refer to Section 7, *Exiting the Config File Manager*.)

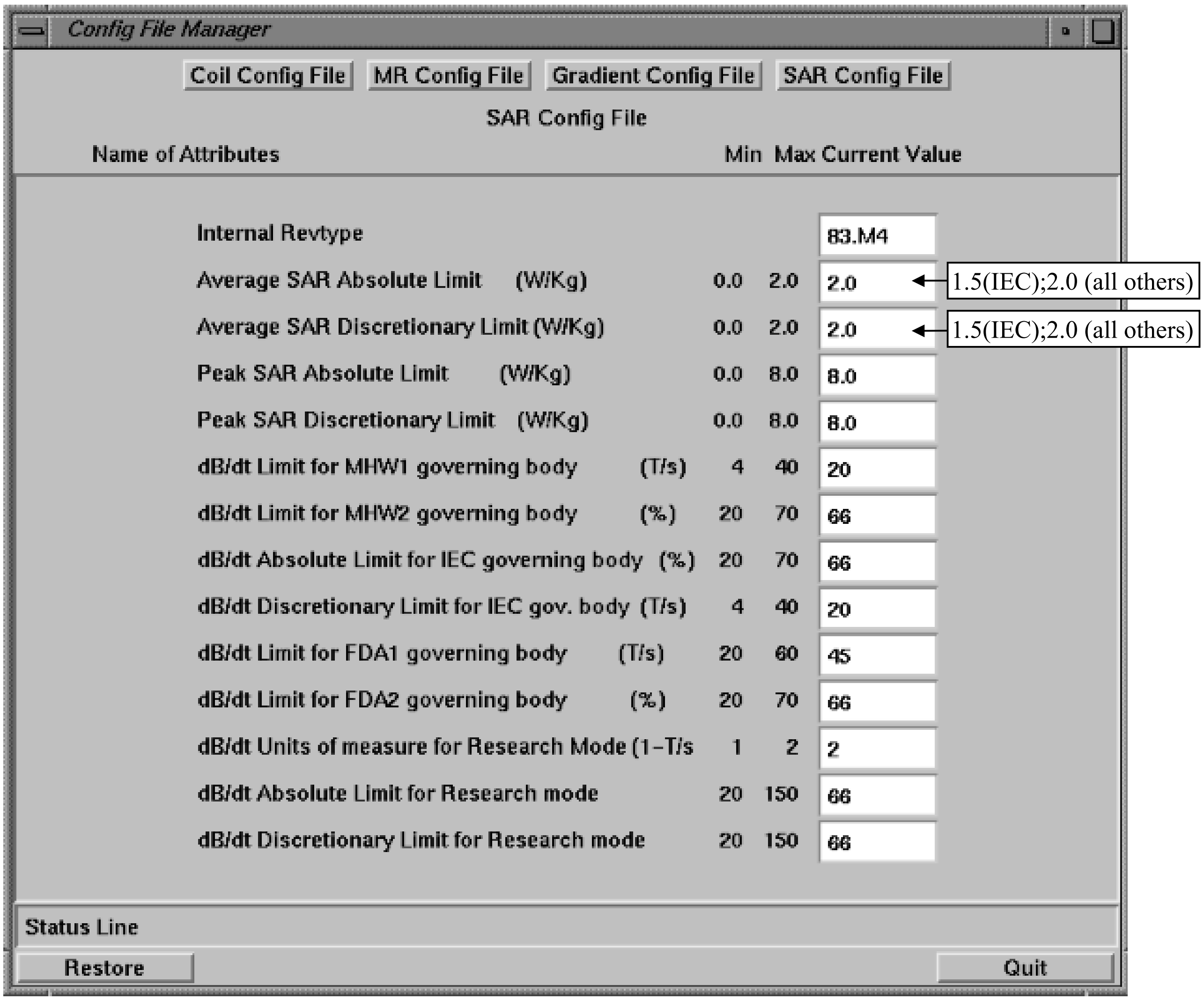
5. See Appendix A for a listing of all the parameters in the Release 9.0 and ASP2 GradientConfig.cfg file.

6- SAR CONFIG FILE

To edit and/or view the values of the SAR Config File, perform the following steps.

1. On the *Config File Manager* main screen, select **[SAR Config File]**. See Illustration 2-1.

2. The *SAR Config File* screen is displayed. See Illustration 6-1.



SAR CONFIG FILE SCREEN (TYPICAL)
ILLUSTRATION 6-1

3. Either click in the fields or tab through the fields to enter values. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located).

4. To apply changes made to the SAR Config File, exit the config file manager. Make sure to click the **[Save]** button when the save dialog box appears. Then, shut down Signa and reboot the computer. (Refer to Section 7, *Exiting the Config File Manager*.)

7- EXITING THE CONFIG FILE MANAGER

1. When finished editing values, click the **[Quit]** button to exit application.

2. Click **[Yes]** in the dialog box question, "*Really Quit?*" See Illustration 7-1.

QUIT DIALOG BOX
ILLUSTRATION 7-1

3. Next, a Save dialog box appears for each configuration file that has changed, (**Example:** The following message will appear, "*Coil Config File has been changed*." To save changes click the **[Save]** button. To not save changes click the **[Do Not Save]** button.

4. To apply changes made in the config files: shut down Signa software and boot computer.
   - Shutting down the software: on the Service Desktop Manager, click on **[System Shutdown]**.
   - Booting the computer: Refer to the procedure for *Bringing the System Up/Down*, Section 5.

APPENDIX A - SITE CONFIGURATIONS/OPTIONS INFORMATION

Site Name:____

Date: ________________

System Configuration Data Sheet (1 of 5)

PARAMETER	DEFAULT [OPTIONS]	SITE VALUE
Internal Rev	SW revision of file	
Line Frequency	60, [50, 60]	
Spectroscopy RF Amplifier	no [yes, no]	
Broadband Transceiver	no [yes, no] yes, for spectroscopy option	
TR Dither	no [yes, no]	
Governing Body	fda2 [iec, mhw1, mhw2, fda2] Selection based on governing body of country where scanner located	
RF Body Coil Vector Z	12431	
RF Body Coil Length	10178	
RF Body Coil Radius	3840	
RF Head Coil Vector Y	191	
RF Head Coil Vector Z	-2800	
RF Head Coil Length	6150	
RF Head Coil Radius	2095	
Head Coil Amp Calibration Factor	[0, xxxxx] 0, for RF cabinets with APB Board or <xxxxx> as determined in procedure for EFB Calibration for RF with EFB Board	
Body Coil Amp Calibration Factor	[0, xxxxx] 0, for RF cabinets with APB Board or <xxxxx> as determined in procedure for EFB Calibration for RF with EFB Board	
Magnet Field Strength	15000 (1.5T) 10000 (1.0T)	
Magnet Serial Number	[xxxxx]	
Room Shielding	no [yes, no] yes, for shielding no, if none	
MagnaShield	no [yes, no] yes, for MagnaShield no, if none.	
Shield Cooler Type	4 [0-6]	
Resistive Shim Power Supply	0 [0-6]	
External Passive Shim	no [yes, no] yes, if external passive shim no, if none.	
Passive Shim Drum	no [yes, no] yes, if passive shim drum no, if none.	
Line Conditioner Type	2 [0-2]	

System Configuration Data Sheet (2 of 5)

PARAMETER	DEFAULT [OPTIONS]	SITE VALUE
Isocenter Vector Along Z-axis	Initial Values: 8914(1.5T) 12431 (1.0T) 8981 (Cx Magnet) 12520 (Non-Magnishield Magnet) 15250 (Magnishield Magnet) (Final values determined in procedure for Electrical Isocenter Calibration)	
Table Movement Limit	31750	
Beginning of Travel (mm)	50	
End of Travel (mm)	2568 If cradle travel has not been calibrated then set to 2550 mm. Refer to procedure for Longitudinal Drive Calibration.	
Max Grad Shim Value	280	
Head Plate Dissipation Limit	800	
Body Plate Dissipation Limit	5000	
Head Power Supply Limit (watts)	350	
Body Power Supply Limit (watts)	4000	
Head Plate Voltage	3000	
Body Plate Voltage	6000	
Head Quiescent Plate Current	.15	
Body Quiescent Plate Current	.7	
Head Load Line Resistance	857	
Body Load Line Resistance	394	
Network Loss Value	1.0715	
Fixed Hardware Trigger Delay	10	
Additional CINE Filter Trigger Delay	28	
Minimum Bandwidth for VBI	2	
Maximum Bandwidth for VBI for Receiver 1 (*Use Guided Install Hardware tab to change value*)	[62.5, 125] 62.5, for 62.5 KHz Bandwidth Receivers 125, for 100 KHz Bandwidth Receivers	
Maximum Bandwidth for VBI for Receiver 2	(same as Receiver 1)	
Maximum Bandwidth for VBI for Receiver 3	(same as Receiver 1)	
Maximum Bandwidth for VBI for Receiver 4	(same as Receiver 1)	
Extended Dynamic Range Threshold	63	
CERD Pre-Amp Warm Up Time (s) *(this parameter is not in ASP2 SW)*	900	
CERD ADC Warm Up Time(s) *(this parameter is not in ASP2 SW)*	1200	
CERD Ov Osc Warm Up Time(s) *(this parameter is not in ASP2 SW)*	300	
Multi Coil T/R Driver Brd Present	yes [yes, no]	
Header Body Single Line Present	yes [yes, no]	

System Configuration Data Sheet (3 of 5)

PARAMETER	DEFAULT [OPTIONS]	SITE VALUE
Bore Monitor Level 1 Trip Point	29 [28-31] 25, for TwinSpeed 28, CV/i with ACGD Wide Open Enclosures 31, for CV/i with SGD Wide Open and all MR/i Wide Open Enclosures 29, for other Enclosures	
Bore Monitor Level 2 Trip Point	33 [29-36] 29 for TwinSpeed 32, CV/i with ACGD Wide Open Enclosures 36, for CV/i with SGD Wide Open and all MR/i Wide Open Enclosures 33, for other Enclosures	
Hysteresis Value For Bore Temp Mon	[0-3] 3, for TwinSpeed 3, for Wide Open Enclosures 2, for other Enclosures	
Offset Value For Bore Temperature	0 [-9 to 9]	
Rated Output Power (body), Watts	16000 (1.5T) 7500 (1.0T)	
Rated Output Power (head), Watts	2000 (1.5T) 750 (1.0T)	
Max Average Power (body), Watts	1000 (1.5T) 750 (1.0T)	
Max Average Power (head), Watts	100 (1.5T) 75 (1.0T)	
Max Pulse Energy (body), Joules	60.0 (1.5T) 30.0 (1.0T)	
Max Pulse Energy (head), Joules	6.0 (1.5T) 3.0 (1.0T)	
Max Pulse Width (body), ms	20 (1.5T) 30 (1.0T)	
Max Pulse Width (head), ms	20 (1.5T) 30 (1.0T)	
Max Duty Cycle (body)	0.60	
Max Duty Cycle (head)	0.60	
RF Amplifier Type	0 [0,1,3,4] 0, Erbtec Tube 3, Teal Solid State (1.0T) 4, Analogic Solid State (1.5T) 10, SRFD2 Module (1.5T)	
Power Monitor Type	[0,1] 0 = for 1.0T 1 = for 1.5T	
Image Compression Value	73 [50-100]	
Gradient GAP or GIP Board Present	yes [yes, no] yes, for ACGD yes, for 8645/GRAM yes, for SGD Hi Slew no, for SGD Base	
Gradient Amplifier's Type	[8280, 8645, 8651, 8915] 8915, for ACGD 8645, for 8645 amplifiers 8651, for SGD HiSpeed or EchoSpeed 8280, for SGD Base	

System Configuration Data Sheet (4 of 5)

PARAMETER	DEFAULT [OPTIONS]	SITE VALUE
X-axis GRAM	gram-1 [no-gram, gram-0 gram-1, gram-2] no-gram, for Base gram-1 for all others	
Y-axis GRAM	(Same as X-axis GRAM)	
Z-axis GRAM	(Same as X-axis GRAM)	
X-axis ASM Present *(Only for 8645 amp)*	yes [yes, no] yes or no, for SGD or ACGD yes, for 8645	
Y-axis ASM Present *(Only for 8645 amp)*	(Same as X-axis ASM)	
Z-axis ASM Present *(Only for 8645 amp)*	(Same as X-axis ASM)	
X-axis GASM Present *(Only for 8645 amp)*	yes [yes, no] yes or no, for SGD or ACGD yes, for 8645 HiSpeed or EchoSpeed no, for 8645 Base	
Y-axis GASM Present *(Only for 8645 amp)*	(Same as X-axis GASM)	
Z-axis GASM Present *(Only for 8645 amp)*	(Same as X-axis GASM)	
X-axis Series *(Only for 8645 amp)*	yes [yes, no] yes or no, for SGD or ACGD yes, for 8645 Base or EchoSpeed no, for 8645 HiSpeed	
Y-axis Series *(Only for 8645 amp)*	(Same as X-axis Series)	
Z-axis Series *(Only for 8645 amp)*	(Same as X-axis Series)	
X-axis Number of Power Modules *(Only for 8645 amp)*	[1, 2] 1 or 2, for SGD or ACGD 2, for 8645 Base or EchoSpeed 1, for 8645 HiSpeed	
Y-axis Number of Power Modules *(Only for 8645 amp)*	(Same as X-axis)	
Z-axis Number of Power Modules *(Only for 8645 amp)*	(Same as X-axis)	
Gradient Digital Tuning	yes [yes, no] yes, for HiSpeed, EchoSpeed, or ACGD no, for Base	
Gradient Real Time Monitoring ON	yes [yes, no]	
Gradient Coil Type	[1,2,3] 1 for CRM 2 for BRM 3 for TRM	
maxB1 strength (uTesla)	25.0	
Maximum integrated (B1) ^2 per pulse (uT^2sec)	8.0	
MaxRMS B1 strength for headcoils (uTesla)	7.2	
Faster Prescan on/off	1 [0, 1] 0=Off; 1=On	
CERD Type	001, CERD 002, Universal CERD	
GradCoil *(Not displayed for prior*	2 [1, 2, 6]	

releases, then for ASP2 moved this parameter to GradientConfig.cfg.		

System Configuration Data Sheet (5 of 5)

PARAMETER	DEFAULT [OPTIONS]	SITE VALUE
(All following shaded parameters are present for 9.0 and ASP2. These were primarily added to support HFO systems)		
Software Gradient Update Time	4	
Hardware Gradient Update Time	4	
Software RF Update Time	2	
Hardware RF Update Time	2	
Software SSP Update Time	1	
Hardware SSP Update Time	1	
Scanner Mode	0	
Smart Coil	0	
Table is fixed or not?	N	
Position A, Angle X	0	
Position A, Angle Y	0	
Position A, Angle Z	0	
Position B, Angle X	0	
Position B, Angle Y	0	
Position B, Angle Z	0	
Position C, Angle X	0	
Position C, Angle Y	0	
Position C, Angle Z	0	
Offset X	0	
Offset Y	0	
Offset Z	0	
Does site have moving metal compensation	0	
TG of moving metal coil	200	
R1 of moving metal coil	13	
DFM delta frequency	0	
Idle time after scan until coldhead operation is restored	30	
X shim for moving metal small sample	0	
Y shim for moving metal small sample	0	
Z shim for moving metal small sample	0	
Z2 shim for moving metal small sample	0	
Flag to indicate if system has a Z2 coil	0	
FFT coefficient for spiral (128 matrix)	0.0006555	
FFT coefficient for spiral (256 matrix)	0.0006555	
FFT coefficient for spiral (512 matrix)	0.0006555	
FFT coefficient for spiral (1024 matrix)	0.0006555	
Spiral regridding coefficient	0.00001035	
Gradient amplifier group delay	120	
RF Delay	56	
Stand By Time	60	
DBdtDistance	27.5	
Digital tuning board type	Yes	

Scale used for multiplying DFM shim current	300	
DBdtLimit	0	

APPENDIX A - SITE CONFIGURATIONS/OPTIONS INFORMATION (continued)

Site Name:____
Date: __________________
Gradient Configuration Data Sheet, Page 1 of 2

For **TwinSpeed,** there are two gradient configuration files: GradientConfig.WHOLE and GradientConfig.ZOOM.

PARAMETER	DEFAULT	SITE VALUE
File format revision number	(Revision number of file)	
X-Axis Gradient Setting to Obtain cfxfs	30000 (ACGD, SR77/120) 25000 (ACGD, TwinSpeed) 20000 (ACGD SR50) 31000 (GRAM, Base) 29200 (SGD with CRM) 25165 (SGD with BRM)	
Y-Axis Gradient Setting to Obtain cfyfs	same as X-Axis	
Z-Axis Gradient Setting to Obtain cfzfs	same as X-Axis	
X-axis Gradient Delay *(for CRM systems beginning at Release 8.4)*	180.0 (CRM)	
Y-axis Gradient Delay *(for CRM systems beginning at Release 8.4)*	180.0 (CRM)	
Z-axis Gradient Delay *(for CRM systems beginning at Release 8.4)*	180.0 (CRM)	
(All following shaded parameters are present for 9.0 and ASP2)		
Z2-axis Gradient Setting to obtain cfzfs	15839	
Gradient Configuration Mode: SR	17 (8645 Base) 20 (SGD Base) 77 (HiSpeed & Twin) 120 (EchoSpeed)	
Gradient coil type	2 (BRM coil) 3 (TRM coil) 6 (CRM coil)	
Patient Weight Limit (Lbs)	350	
Gradient Module Power *(ACGD & SGD only)*	6300	
Lcoil	1.40	
Pmgs	3000	
Rac	152	
Rcoil *(ACGD & SGD only)*	0.35	
Rheadroom *(ACGD & SGD only)*	0.10	
SRMode20T	33.6 (for SR120, 77, 50) 20.0 (for SR20)	
SRModeConv	77.0 (for SR120, 77, 50) 20.0 (for SR20)	
Sdlim	0.60	
Vheadrooom *(ACGD & SGD only)*	17	
MGRAMpower	9000 (for SR120) 6300 (for SR77, 50, 20)	
MaxB1RMS	3.6	
Gradient group delay for X axis	180.000000	
Gradient group delay for Y axis	180.000000	

Gradient Configuration Data Sheet, Page 2 of 2

Gradient group delay for Z axis	180.000000	
Gradient Group Delay for 0.2T	0	
Gradient Group Delay for 0.35T	368	
Gradient Group Delay for 0.5T	0	
Gradient Group Delay for 0.7T	200	
Gradient Group Delay for 1.0T	See Table A-1	
Gradient Group Delay for 1.5T	See Table A-1	
RF Delay for 0.2T	0	
RF Delay for 0.35T	368	
RF Delay for 0.5T	0	
RF Delay for 0.7T	220	
RF Delay for 1.0T	See Table A-1	
RF Delay for 1.5T	See Table A-1	
X MiniGram Power	1500	
xRsat *(ACGD & SGD only)*	0.022	
xVpwm *(ACGD & SGD only)*	24.0 (for SR77/SR50) 35.0 (for SR120)	
xVsat *(ACGD & SGD only)*	4.8	
YMGRAMpower *(ACGD & SGD only)*	1500	
yRsat *(ACGD & SGD only)*	0.022	
yVpwm *(ACGD & SGD only)*	24.0 (for SR77/SR50) 35.0 (for SR120)	
yVsat *(ACGD & SGD only)*	4.8	
ZMGRAMpower *(ACGD & SGD only)*	1500	
zRsat *(ACGD & SGD only)*	0.022	
zVpwm *(ACGD & SGD only)*	24.0 (for SR77/SR50) 35.0 (for SR120)	
zVsat *(ACGD & SGD only)*	4.8	

TABLE A-1
RF DELAY & GRADIENT GROUP DELAY 1.5T & 1.0T

Configuration	Gradient Delay 1.0T	RF Delay 1.0T	Gradient Delay 1.5T	RF Delay 1.5T
ACGD SR120	0	0	176	164
ACGD SR77	176	164	176	164
ACGD SR50	168	160	168	160
SGD SR120	0	0	176	164
SGD SR77	176	164	176	164
SGD SR20	220	204	220	204
8645 SR120	0	0	176	160
8645 SR77	172	158	172	156
8645 SR20	184	176	184	176

APPENDIX B - COIL DESCRIPTION INFORMATION

This appendix contains three tables of coil descriptions, refer to the appropriate table for your system:

- Table 1 – 1.5T Coils (BRM)
- Table 2 – 1.0T Coils
- Table 3 - 1.5T Coils (CRM)

Table 1: 1.5T Coils (BRM)

Legend: New names (since 8.2.5) are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires "N Coil Recon." ♦ configuration benefits from "N Coil Recon."

Coil Name	Manufacturer	Comments
HEAD	GE	no comment
BODY	GE	no comment
EXTREM	GE	original linear extremity coil, transmit/receive
KNEEPA	Medrad	four element phased array
QUADKNEE	Medical Advances	single channel quadrature knee, transmit/receive
5GP	GE	single channel receive only
3INCH	GE	single channel receive only
ANTNECK	Medrad	linear single channel
POSTNECK	Medrad	single channel receive only, obsolete
QUADCSP	Medrad	single quadrature cervical spine coil
QUAD T/L	Medrad	single channel quadrature receive only
USCTLTOP	USAI	quadrature 6 element phased array
USCTLMID	USAI	quadrature 6 element phased array
USCTLBOT	USAI	quadrature 6 element phased array
USCS123 ♦	USAI	quadrature 6 element phased array
USCT1234 ♣	USAI	quadrature 6 element phased array
USTL345 ♣	USAI	quadrature 6 element phased array
USLS456 ♦	USAI	quadrature 6 element phased array
USCS12	USAI	quadrature 6 element phased array
USTS23	USAI	quadrature 6 element phased array
USTS34 ♣	USAI	quadrature 6 element phased array
USTL45	USAI	quadrature 6 element phased array
USLS56	USAI	quadrature 6 element phased array
CTLTOP	GE by Teledyne	linear 6 element phased array
CTLMID	GE by Teledyne	linear 6 element phased array
CTLBOT	GE by Teledyne	linear 6 element phased array
CS123 ♦	GE by Teledyne	linear 6 element phased array
CT234 ♣	GE by Teledyne	linear 6 element phased array
TL345 ♣	GE by Teledyne	linear 6 element phased array
LS456 ♦	GE by Teledyne	linear 6 element phased array
CS12	GE by Teledyne	linear 6 element phased array
TS23	GE by Teledyne	linear 6 element phased array
TS34 ♣	GE by Teledyne	linear 6 element phased array
TL45	GE by Teledyne	linear 6 element phased array
LS56	GE by Teledyne	linear 6 element phased array
PELVIC	Medrad	uses Medrad ATD3 with endorectal probe and pelvic array
ENDOREC	Medrad	requires Medrad ATD2 with stand-alone endorectal probe
CARDIAC	GE	four channel phased array
TORSOPA	GE	new "TORSOPA" array replaces "TORSO" - configuration is different!
TORSO	GE by Gore	Original 1.5T "TORSO" array, now obsolete
BREAST	Medrad	single channel quadrature configuration

Table 1: 1.5T Coils (BRM)		
Legend: New names (since 8.2.5) are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires "N Coil Recon." ♦ configuration benefits from "N Coil Recon."		
Coil Name	**Manufacturer**	**Comments**
BREASTPA	Medrad	phased array configuration
R_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the right breast only, using two phased array channels.
L_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the left breast only, using two phased array channels.
2_BREAST	MRI Devices	Phased array Open Breast Coil configured to image both breasts, using four phased array channels.
SHOULDER	Medrad	linear single channel
SHOPA2	Medrad	two element phased array
SHOPA3 ♦	Medrad	three element phased array
SHLDRPA4	MRI Devices	shoulder four channel phased array, called 4SHOULDR in MRI Devices service documentation
DUAL	GE by Teledyne	2 channel PA adapter for use with pairs of identical single channel coils
DUALFLEX	GE	two body flex coils used with "DUAL" adapter to create two channel PA
BODYFLEX	GE	single body flex coil used with fast receiver, EPI only
TMJPA	GE by Teledyne	two three inch GP coils permanently attached to a DUAL adapter
GPFLEX	Gore	linear single channel, may be some old equivalent GE-built versions in installed base
NEUROVAS	Medrad	single transmit receive neurovascular coil
NVHEAD	MRI Devices	single channel phased array configuration works with conventional receiver for high SNR head scans
NVARRAY	MRI Devices	four channel phased array configuration
NVFASTHD	MRI Devices	Previously called "FASTHEAD" - single channel phased array configuration works with fast receiver for functional imaging
P31_FLEX	GE by USAI	Single channel , transmit/receive, phosphorous 31 flex coil for use on systems with multi-nuclear spectroscopy option only
PVUPPER	USAI	Most superior station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.
PVMIDDLE	USAI	Middle station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.
PVLOWER	USAI	Most inferior station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.
TPUPPER *Added @ ASP2*	USAI	Torso-Pelvic Array. This station is used for scanning the abdomen.
TPLOWER *Added @ ASP2*	USAI	Torso-Pelvic Array. This station is used for scanning the pelvis.
TRkneePA *Added @ ASP2*	MRI Devices	A transmit/receive phased array is used for scanning knees and ankles.
TRfootPA *Added @ ASP2*	MRI Devices	A transmit/receive phased array is used for scanning feet.

Table 2: 1.0T Coils		
Legend: New names are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires "N Coil Recon." ♦ configuration benefits from "N Coil Recon."		
Coil Name	**Manufacturer**	**Comments**
HEAD	GE	no comment
BODY	GE	no comment
QUADEXT	GE	original quadrature extremity coil, transmit/receive
QUADKNEE	Medical Advances	single channel quadrature knee, transmit/receive
3INCH	GE	single channel receive only
ANTNECK	Medrad	linear single channel
QUADCSP	Medrad	single quadrature cervical spine coil
QUAD T/L	Medrad	single channel quadrature receive only
USCTLTOP	USAI	quadrature 6 element phased array
USCTLMID	USAI	quadrature 6 element phased array
USCTLBOT	USAI	quadrature 6 element phased array
USCS123 ♦	USAI	quadrature 6 element phased array
USCT234 ♣	USAI	quadrature 6 element phased array
USTL345 ♣	USAI	quadrature 6 element phased array
USLS456 ♦	USAI	quadrature 6 element phased array
USCS12	USAI	quadrature 6 element phased array
USTS23	USAI	quadrature 6 element phased array
USTL34 ♣	USAI	quadrature 6 element phased array
USLS45	USAI	quadrature 6 element phased array
USLS56	USAI	quadrature 6 element phased array
CTLTOP	Medrad	quadrature 7 element phased array
CTLMID	Medrad	quadrature 7 element phased array
CTLBOT	Medrad	quadrature 7 element phased array
CS123 ♦	Medrad	quadrature 7 element phased array
TL345 ♣	Medrad	quadrature 7 element phased array
TL456 ♣	Medrad	quadrature 7 element phased array
LS567 ♣	Medrad	quadrature 7 element phased array
CS12	Medrad	quadrature 7 element phased array
TS34 ♣	Medrad	quadrature 7 element phased array (originally called TL34 in early 8.2.5)
TL45	Medrad	quadrature 7 element phased array (originally called TS45 in early 8.2.5)
LS56	Medrad	quadrature 7 element phased array
LS67	Medrad	quadrature 7 element phased array
PELVIC	Medrad	uses Medrad ATD3 with endorectal probe and pelvic array
ENDOREC	Medrad	requires Medrad ATD2 with stand-alone endorectal probe
CARDIAC	GE	four channel phased array
TORSOPA	GE	new "TORSOPA"
TORSOPA	GE by Gore	Original 1.0T "TORSOPA" array, Japan only, now obsolete
BREAST	Medrad	single channel quadrature configuration
R_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the right breast only, using two phased array channels.
L_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the left breast only, using two phased array channels.
2_BREAST	MRI Devices	Phased array Open Breast Coil configured to image both breasts, using four phased array channels.
SHOULDER	Medrad	linear single channel
SHLDRPA4	MRI Devices	shoulder four channel phased array, called 4SHOULDR in MRI Devices service documentation
DUAL	GE by Teledyne	2 channel PA adapter for use with pairs of identical single channel coils

Table 2: 1.0T Coils		
Legend: New names are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires "N Coil Recon." ♦ configuration benefits from "N Coil Recon."		
Coil Name	**Manufacturer**	**Comments**
DUALFLEX	GE	two body flex coils used with "DUAL" adapter to create two channel PA
TMJPA	GE by Teledyne	two three inch GP coils permanently attached to a DUAL adapter
GPFLEX	Gore	linear single channel, may be some old equivalent GE-built versions in installed base
NVHEAD	MRI Devices	single channel phased array configuration works with conventional receiver for high SNR head scans
NVARRAY	MRI Devices	four channel phased array configuration
PVUPPER	USAI	Most superior station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.
PVMIDDLE	USAI	Middle station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.
PVLOWER	USAI	Most inferior station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.

Table 3: 1.5T Coils (CRM)

Legend: New names (since 8.2.5) are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires "N Coil Recon." ♦ configuration benefits from "N Coil Recon."

Coil Name	Manufacturer	Comments
HEAD	GE	no comment
BODY	GE	no comment
EXTREM	GE	original linear extremity coil
KNEEPA	Medrad	four element phased array
QUADKNEE	Medical Advances	single channel quadrature knee, transmit/receive
5GP	GE	single channel receive only
3INCH	GE	single channel receive only
ANTNECK	Medrad	linear single channel
POSTNECK	Medrad	single channel receive only, obsolete
QUADCSP	Medrad	single quadrature cervical spine coil
QUAD T/L	Medrad	single channel quadrature receive only
USCTLTOP	USAI	quadrature 6 element phased array
USCTLMID	USAI	quadrature 6 element phased array
USCTLBOT	USAI	quadrature 6 element phased array
USCS123 ♦	USAI	quadrature 6 element phased array
USCT234 ♣	USAI	quadrature 6 element phased array
USTL345 ♣	USAI	quadrature 6 element phased array
USLS456 ♦	USAI	quadrature 6 element phased array
USCS12	USAI	quadrature 6 element phased array
USTS23	USAI	quadrature 6 element phased array
USTL34 ♣	USAI	quadrature 6 element phased array
USLS45	USAI	quadrature 6 element phased array
USLS56	USAI	quadrature 6 element phased array
CTLTOP	GE by Teledyne	linear 6 element phased array
CTLMID	GE by Teledyne	linear 6 element phased array
CTLBOT	GE by Teledyne	linear 6 element phased array
CS123	GE by Teledyne	linear 6 element phased array
CT234 ♣	GE by Teledyne	linear 6 element phased array
TL345 ♣	GE by Teledyne	linear 6 element phased array
LS456	GE by Teledyne	linear 6 element phased array
CS12	GE by Teledyne	linear 6 element phased array
TS23	GE by Teledyne	linear 6 element phased array
TL34 ♣	GE by Teledyne	linear 6 element phased array
LS45	GE by Teledyne	linear 6 element phased array
LS56	GE by Teledyne	linear 6 element phased array
PELVIC	Medrad	uses Medrad ATD3 with endorectal probe and pelvic array
ENDOREC	Medrad	requires Medrad ATD2 with stand-alone endorectal probe
CARDIAC	GE	four channel phased array
TORSOPA	GE	new "TORSOPA" array replaces "TORSO" - configuration is different!
TORSO	GE by Gore	Original 1.5T "TORSO" array, now obsolete
BREAST	Medrad	single channel quadrature configuration
BREASTPA	Medrad	phased array configuration
R_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the right breast only, using two phased array channels.
L_BREAST	MRI Devices	Phased array Open Breast Coil configured to image the left breast only, using two phased array channels.
2_BREAST	MRI Devices	Phased array Open Breast Coil configured to image both breasts, using four phased array channels.
SHOULDER	Medrad	linear single channel

Table 3: 1.5T Coils (CRM)		
Legend: New names (since 8.2.5) are in bold. Coils that were on the original list whose names have been changed are in bold italics. ♣ configuration requires “N Coil Recon.” ♦ configuration benefits from “N Coil Recon.”		
Coil Name	**Manufacturer**	**Comments**
SHOPA2	Medrad	two element phased array
SHOPA3 ♦	Medrad	three element phased array
SHLDRPA4	MRI Devices	shoulder four channel phased array, called 4SHOULDR in MRI Devices service documentation
DUAL	GE by Teledyne	2 channel PA adapter for use with pairs of identical single channel coils
DUALFLEX	GE	two body flex coils used with “DUAL” adapter to create two channel PA
BODYFLEX	GE	single body flex coil used with fast receiver, EPI only
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PVLOWER	USAI	Most inferior station of the twelve element phased array Peripheral Vascular Array. This station uses four phased array channels, two in the posterior array and two in the anterior array.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	May 5, 1998	K. Keshena	Preliminary version.
0	July 7, 1998	M. Keber	Miscellaneous updates based on validation.
1	Oct 14, 1998	M. Keber	Updated for program change from Release “CV1” to “8.2.5”.
2	Dec 1, 1998	M. Keber	Updated for 8.2.5 M3 changes. Added step in Section 3-1 to set Number of Fast Receivers = 1 for EchoSpeed systems.
3	Dec. 22, 1998	K. Keshena	Added new option for RF Amplifier Type in Appendix A, page 2 of 3.
4	Feb 12, 1999	M. Keber	Added default value for “Multi Coil Recon Enable” parameter in section 3.4 for 8.2.5 software. Delete key can add bad characters when editing fields.
5	May 23, 1999	S.M. Atladottir	Added additional parameter information.
6	Jul 2, 1999	G. Boerner	Edits based on 8.3 Tools Validation. Also added notes in Coil section that with 8.3 all coils are configured with software load and any coils not being used will need to be deleted.
7	Oct. 13, 1999	J. Wolak	Added new MR Config File parameters for Release 8.3 and cleaned up Appendix A; other 8.3 notes.
8	Mar. 1, 2000	J. Wolak	Updates for 8.3 backspace fix and other minor changes. Added Appendix B; coil definitions
9	Sept 22, 2000	M. Keber	Updated Gradient Config screen and data sheet for Release 8.4 changes; updated for new backspace problem beginning with 8.3.
10	Jan. 22, 2001	J. Wolak	Updated procedure for ASP2 impact. Many new parameters in MRconfig.cfg & GradientConfig.cfg files.
11	March 22, 2001	M. Jones	Added “(TYPICAL)” to Ill. 2-1, 3-1, 3-2, 3-3, 4-1, and 6-1 titles to allow for software differences. Deleted old software reference from Apx. A, sheet 2. Other very minor corrections.
12	Aug. 8, 2001	J. Wolak	Updates for TwinSpeed for Release 9.0 by merging in Jeremy Gerber’s changes for TwinSpeed (from his rev 9 and 10 versions). Also, added configuration file parameter values for TwinSpeed systems in Appendix A.
13	Sept. 21, 2001	K. Keshena	Updated System Configuration Data Sheet (3 of 5) in Appendix A. to include the default parameters of Bore Monitor Level Trip Points for CV/i.
14	July 30, 2002	K. Keshena	Added new option for RF Amplifier Type for the SRFD2 1.5T module in Appendix A, page 3 of 5.
15	Feb 17, 2003	P. Kargard	Added information concerning autoshim feature.